

Health & Safety

Standard Lock out / Tag out



Reference

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Table of contents

1. Objective	2
2. Scope	2
3. Reference documents	2
4. Abbreviations, terms and definitions	2
5. Roles and responsibilities	3
5.1. Operations / Production manager	3
5.2. Isolating Authority / Consignment Officer	4
5.3. Intervention officer	4
5.4. Works officer / Operations officer	4
5.5. Performer	4
6. Training, competencies and authorizations	5
6.1. Personnel involved in electrical work	5
6.2. Personnel involved in other activities under isolation	6
7. Isolation preparation	6
7.1. Design of work equipment	6
7.2. Risk analysis	6
7.3. Daily briefing	7
8. Perform an isolation	7
8.1. Types of electrical lockouts	7
8.2. Recording steps and tasks to carry out	7
8.3. Logging equipment	8
8.4. Logging process	9
9. Related documents	9

1. Objective

The objective of this document is to describe the measures to be taken by Bouygues Travaux Publics (BYTP) to ensure that Lock-out / Tag-out process (LOTO) is properly implemented before any preventive or corrective maintenance, set-up adjustment or reparation intervention, in order to prevent any risk of exposure to an energy source or release of energy or untimely movement. This includes all forms of energy (electrical, hydraulic, pneumatic, mechanical, etc.).

System testing, machines start-up, and commissioning activities are not within the scope of this document.

2. Scope

This document applies to all companies working on sites (including service providers and subcontractors).

In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

The requirements related to the management of interfaces with existing networks, buried or aerial, are defined in the **Standard Construction machinery and associated maintenance [BYTP-REF-H&S-2215]**.

3. Reference documents

- NF C18-510: "Operations on electrical works and installations and in an electrical environment – Prevention of electrical risk"
- NF EN ISO 12100: "Safety of machines General design principles Risk assessment and risk reduction"

4. Abbreviations, terms and definitions

Term / Abbreviation	Description
ELV, LV	Extra Low Voltage, Low Voltage
HT	High tension
VAT	Voltage Absence Tester
GF	Ground Fault
Intervention	General LV interventions are limited by the physical characteristics of the circuits on which they are authorized. They are reserved for circuits: • Powered by LV or ELV • Protected against short circuits by a current protection device rated at or below 63 A, in alternating current, and at or below 32 A, in direct current

Authorization – Authorized person	Recognition by the employer of the ability of a person placed under their authority to perform the tasks entrusted to them safely in relation to electrical and mechanical risks (machines and fluid networks)
Operation	Activity carried out, either directly on the equipment or in the vicinity of electrical, mechanical, or network equipment.
Lock out	The placement of a lock out device on an energy isolating device in accordance with an established procedure, ensuring that the energy isolating device and the equipment being controlled cannot be operated until the lock out device is removed.
Reinstatement	Set of provisions making it possible to restore previously Locked out equipment to operating condition, while ensuring the safety of people and equipment. The reinstatement procedure specifies the composition and order of the steps based on a risk analysis and testing the modifications implemented.
Prior check	Verification carried out before equipment lock-out. Type of prior checks: • Information for operators on the locking out of the equipment • Intervention area(s) signaled and barricaded if required • Identification of the parts of equipment(s) which remain energized (partial lockout) Absence of interaction between the different parts of equipment(s) (recorded parts, non-recorded parts).
Lock-out device	Physical device or mechanism used to disable machinery or equipment to prevent the release of hazardous energy. The different existing devices are described in §8.3
Machine	All sets of mechanical elements which can be in motion, fast or slow, rectilinear or circular, continuous or reciprocating, including construction machinery.

5. Roles and responsibilities

5.1. Operations / Production manager

- He organizes and coordinates activities on the works
- He designates the Isolating Authority(ies)
- It validates the lock-out/tag-out procedure for the site
- It validates requests for lock-out and reinstatement in the event of preventive maintenance, corrective maintenance or requests them in the event of emergency intervention, breakdowns, etc.

5.2. Isolating Authority / Consignment Officer

- He is responsible for operating acts to provide the safe conditions allowing stakeholders to carry out work on a designated equipment
- He has a very good knowledge of the related equipment
- He carries out the lock-out and reinstatement needed for the operations on installations defined through his competency's authorization
- He ensures a preliminary check before any activity
- He issues the corresponding isolated certificate
- He stops the activity in case of unforeseen event/condition

5.3. Intervention officer

- He is responsible for carrying out low voltage (LV) interventions (troubleshooting, corrective maintenance, short-term and unplanned activities, etc.).
- He has the ability to analyze and sufficient knowledge of the operation of the installation or electrical equipment on which he intervenes
- He performs a preliminary check before any activity
- He carries out himself the lockout/tagout procedures for his LV intervention
- He stops the activity when he detects an unforeseen event/condition

5.4. Works officer / Operations officer

- He is responsible for ensuring the effective management of the works and for taking or ensuring the necessary measures to ensure his own safety, that of the personnel under his command, as well as that of third parties
- He is in contact with the Isolating Authority, reporting on the state of lock-out or reinstatement
- He leads the briefing at the beginning of the shift and the debriefing at the end of the shift
- He ensures that the work has been clearly defined and familiarizes himself with the content of the activity file.
- He checks that there are no risks not taken into account
- He ensures that the workers have the appropriate authorizations for the work
- He collects the certificate of lock-out from the Isolating Authority
- He respects the different stages of lock-out procedure which are under his responsibility
- He identifies the equipment related to the operation to be performed and verifies all the key points
- When possible, he places his own padlock on the isolation device or on the key lock-out box.
- He ensures a preliminary check before any activity
- He stops the activity when he detects an unforeseen event/condition

5.5. Performer

- He carries out the operations while respecting the instructions imposed by the Works officer / Operations officer / Intervention officer
- He places his own padlock on the isolation device or on the key lock-out box, if deemed necessary and possible during the operation risk analysis.
- He ensures a preliminary check before any activity
- He stops the activity when he detects an unforeseen event/condition

6. Training, competencies and authorizations

6.1. Personnel involved in electrical work

NF C 18-510 defines the minimum required skills based on:

- Role in electrical activities
- Technical skill level
- Voltage level(s) concerned: mainly in our activities: LV (Low Voltage) and HV (High Voltage up to 50kV)

Authorized people benefit from qualifying training and retraining every 3 years.

Operations manager : an operations manager is identified in the organization chart of the site , or of the organizational structure on which the site depends . It is attached to the delegatee of power / Project Director or this structure.

Alternating current / Direct Current Isolating Authority: the Isolating Authority is trained and holds an authorization. Under the authority of the operations manager

Junction intervention manager: the intervention manager is trained and holds an authorization. He receives an intervention authorization issued by the head of the establishment or the operations manager or be authorized to access the installation by the person responsible for the installation authorizing access.

Works manager high voltage / low voltage: the works manager is trained and holds an authorization. He carries out the operations entrusted to him and in particular, those that are specified in the certificate of lock out for work or in the certificate of first stage of lock out.

Executor(s) high voltage / low voltage: The executor is trained and holds an authorization. He ensures the execution of operations. He operates under the supervision of a works manager or an intervention manager.

Other non-electrical authorized person: Trained and holder of an authorization. He carries out non-electrical works. He works in an area affected by lockout, is informed of lockout procedures so that he does not attempt to remove lockout/lockout devices.

6.2. Personnel involved in other activities under isolation

Depending on the risk analyzes of a project's activities, complementary skills to those of electrical lock-out are required:

- Hydraulic
- Mechanical
- Pneumatic

The personnel responsible for carrying out these isolations are trained and authorized.

7. Isolation preparation

7.1. Design of work equipment

Work equipment are designed and constructed to ensure that their installation and maintenance, in normal conditions, do not expose people to a risk of harm to their health or safety. Work equipment subject to Machinery Directive 2006/42/EC is systematically equipped with devices such as a padlockable disconnecter, padlockable purge valve, etc. (Reference **Integrating Safety in Design into technical studies [TEC PRO 5604]**).

7.2. Risk analysis

A risk analysis is performed before any operation in order to define and implement, during operations, appropriate prevention measures for the protection of people and property. It is carried out before each phase of work and takes into account the area where people and tools move during work. This zone must be perfectly delimited in space and defined in time. This risk analysis is carried out jointly by the Isolating Authority , the Works officer with a view to ensuring his safety, that of his team and of third parties, and each performer with a view to ensuring his own safety and that of the people impacted.

The analysis is carried out taking into account in particular the risks presented by:

- Equipment characteristics
- Considered Method statement
- The volume actually occupied by operators, including in their travel
- Normal gestures and reflexes of operators
- Possible movements of equipment , possible falls of tools, materials, parts or machines used for the planned operations and the equipment and coactivity (presence of other activities nearby, other workstations remaining in operation)

The results of the risk analysis is used to specify the following provisions in the project related documentation:

- The delimitation of intervention zones, supervised by a single person responsible for coordinating current operations
- Systematic information for operators on planned operations
- The designation of the Isolating Authority and personnel involved (internal or external)
- Coordination of lock-out and reinstatement follow-ups in the event of a change of work team (for example, work in successive shifts or work over several days)
- Taking into consideration the vicinity of the installation (existing networks under tension, under pressure, etc.) for the lock-out

7.3. Daily briefing

A briefing is systematically carried out at the start of the shift, led by the Works officer / Operations officer. It is the opportunity to:

- Consider the risks in perspective of the planned activities.
- Know the roles and responsibilities of each person involved in these activities.
- Prepare individually and collectively for action, to anticipate the management of possible problems and their solutions.

8. Perform an isolation

8.1. Types of electrical lockouts

For electrical energy, there are two types of lockouts:

- Lock out in 1 step: pre-identification and all LOTO operations are carried out by the Isolating Authority.
- Lock out in 2 stages: the Isolating Authority only carries out the pre-identification and the LOTO operations: separation and condemnation. It is the works manager who carries out the identification, No Voltage Checker and short circuit Grounding.

The recording steps and methods are described in the document **Lockout Steps [BYTP-INF-H&S-21691]**.

8.2. Recording steps and tasks to carry out

The lock out of a work equipment takes into account all the different energies present (electrical, mechanical or fluidic) while respecting a lock out order. The LOTO procedure includes phases whose order and execution may be modified after the risk analysis depending on the specificity of the case considered:

- Separation
- Condemnation and signaling
- Dissipation or retention/containment, (concerns mechanical and fluidic energies)
- Identification
- Verification of absence of energy
- Grounding and Short Circuit (concerns electrical energy to ensure any inadvertent re-energization)

Refer to the document **Lockout Steps [BYTP-INF-H&S-2169]** which details the practical tasks/operations to be carried out for each of the lockout areas: electricity/mechanics (machines stationary and moving) and pneumatic/hydraulic.

8.3. Logging equipment

Lockout equipment	Elements locking the locking devices to physically block the equipment. These include lockout padlocks, lock out jaws and cables, lockout accessories for valves, pneumatic fittings, electrical outlets or circuit breakers.
Lockout box	After locking the equipment, the locking key(s), for starting, locking, etc. of the equipment are placed into a lockout box. Each concerned workers place their individual padlock on the box. It Allows to ensure that no one has access to the keys located in the box, unless each participant has removed his personal padlock.
Multiple safety jaws	Allows multiple stakeholders to condemn a single energy source.
Lockout padlock	The lockout padlock ensures the closure of the lockout equipment. It is identifiable with a high level of security and solidity. A color-coded system helps identify the type of lockout or company, for example : • Red: electrical lockout • Blue: pneumatic lockout • Green: water deposit • Yellow: gas lockout • Orange: hydraulic lockout • Black: mechanical lockout
Named security padlock	The security padlock consists of a single cylinder guaranteeing optimal security. A unique key is supplied with each padlock and the key will not be removed if the padlock shackle is not properly engaged. Each padlock is customizable (labels specifying ownership <u>and/or user</u>). The color of the nominative padlocks is different from the colors of the lock out padlocks.
Lock out/Tag out device	Safety device to ensure that dangerous machinery is properly shut down and cannot be restarted until maintenance or repair work is completed. The isolated power sources are then locked and a tag is placed on the lock identifying the operator who placed it. The operator then retains the key to the lock, ensuring that he is the only person who can remove the lock and start the machine – The use of a lockout box is recommended.

The lockout devices and materials are detailed in the document **Lockout Devices and Materials [BYTP-INF-H&S-2031]** .

8.4. Logging process

The deposit arrangements are detailed in the documents:

- **Logging flowchart [BYTP-INF-H&S-2148]**
- **Long-term lockout principle diagram [BYTP-INF-H&S-2149]**
- **Request – lockout certificate – Unlock out [BYTP-FOR-H&S-2150]**

Examples of good practices in terms of lockout are available in the Document **Lock out - Good Practices [BYTP-INF-H&S-2151]**.

9. Related documents

- Annex 01 – Lockout steps (BYTP-INF-H&S-2169)
- Annex 02 – Lockout devices and materials (BYTP-INF-H&S-2031)
- Annex 03 – Lockout flowcharts (BYTP-INF-H&S-2148)
- Annex 04 – Long-term lockout principle diagram (BYTP-INF-H&S-2149)
- Annex 05 – Request–Lockout certificate–Unlock out (BYTP-FOR-H&S-2150)
- Annex 06 – Lock out – Good practices (BYTP-INF-H&S-2151)